

## Homework 5

Week 5

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**Problem 1.** Generate (7,4) Cyclic code from the data (0101, 1001, 0010, 1110, 0000), and its hardware implementation.

### Solution

Take the standard (7, 4) Hamming cyclic code, generated by  $g(x) = x^3 + x + 1$  (binary 1011), so  $n = 7$ ,  $k = 4$ ,  $n - k = 3$ . For a message  $m_3m_2m_1m_0$  write  $m(x) = m_3x^3 + m_2x^2 + m_1x + m_0$ . Systematic encoding shifts the message into the high-order bits and fills the low 3 bits with a parity check:

$$c(x) = x^{n-k}m(x) + r(x), \quad r(x) = x^{n-k}m(x) \bmod g(x),$$

so the codeword is  $[m_3m_2m_1m_0 \mid r_2r_1r_0]$ , and division is mod-2 (XOR instead of subtract/borrow).

**Worked example (0101).**  $m(x) = x^2 + 1$ , so  $x^3m(x) = x^5 + x^3$ . Dividing by  $g(x) = x^3 + x + 1$ :

$$\begin{aligned} x^5 + x^3 &= \underbrace{(x^2)}_{\text{quotient}} (x^3 + x + 1) + \underbrace{x^2}_{\text{remainder}} \\ (x^2)(x^3 + x + 1) &= x^5 + x^3 + x^2 \\ (x^5 + x^3) \text{ XOR } (x^5 + x^3 + x^2) &= x^2 = r(x) \end{aligned}$$

$\deg(x^2) < \deg g(x)$ , so the division stops:  $r = 100$ , giving codeword  $c = 0101\ 100$ . The other four messages divide the same way (each verified independently against the systematic generator matrix built from  $x^{n-k}x^i \bmod g(x)$  for  $i = 0, 1, 2, 3$ ):

| Message $m_3m_2m_1m_0$ | Shifted $x^3m(x)$ | Remainder $r_2r_1r_0$ | Codeword $c$   |
|------------------------|-------------------|-----------------------|----------------|
| 0101                   | 0101000           | 100                   | <b>0101100</b> |
| 1001                   | 1001000           | 110                   | <b>1001110</b> |
| 0010                   | 0010000           | 110                   | <b>0010110</b> |
| 1110                   | 1110000           | 100                   | <b>1110100</b> |
| 0000                   | 0000000           | 000                   | <b>0000000</b> |

**Table 1:** (7, 4) cyclic codewords,  $g(x) = x^3 + x + 1$

**Hardware implementation.** A  $(n, k)$  cyclic encoder is built from an  $(n - k)$ -stage feedback shift register whose taps are the coefficients  $g_0, g_1, g_2$  of  $g(x)$  (here 1, 1, 0;  $g_3 = 1$  is the implicit feedback stage). Let  $b_0, b_1, b_2$  be the register stages and  $e = G \cdot (m \oplus b_2)$  the gated feedback signal, where  $G$  is a control line that is 1 for the first  $k = 4$  clocks and 0 for the last  $n - k = 3$ :

$$b_0 \leftarrow e, \quad b_1 \leftarrow b_0 \oplus e, \quad b_2 \leftarrow b_1.$$

While  $G = 1$  the message bits shift straight through to the output *and* into the register (computing the division above); once all 4 message bits have passed,  $G$  drops to 0, which forces  $e = 0$  (the register now just shifts, no more feedback) and switches the output to  $b_2$ , so  $b_2, b_1, b_0$  – exactly  $r_2r_1r_0$  – shift out over the next 3 clocks.

